

Variants of the alignment problem

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Introduction

- Broadly, there are more problem-driven researchers and more method-driven researchers.
 - A backward-chain-y researcher has interests in some problems and tries to find ways/methods/understanding to solve them.
 - A forward-chain-y researcher has some methods/skills/understanding and tries to develop them and find problems to solve using them.
- [Timothy Gowers makes a similar distinction between problem-solvers and theory-builders in math.](#)
- Alignment is an unusually backward-chain-y field. It is ultimately about solving a specific problem.
- Examples of more problem-driven entities:
 - Paul Erdős, Terry Tao, John von Neumann, Richard Feynman, theoretical CS, Paul Christiano, arguably MIRI
- Examples of more method-driven entities:
 - Grothendieck, Bourbaki, category theory, Dirac, SLT alignment people, infrabayesianism, maybe: SAE interp, ARC, Redwood
- This talk will be a problem-first view of the field of alignment.

What should we mean by “the alignment problem”?

- There's this worrying situation that AI might cause the future to be bad/empty.
- Could we just say that the alignment problem is the problem of making it so this does not happen?
 - This seems too inclusive, as then banning AI would be a possible solution to the alignment problem. While I have nothing against banning AI, it's weird to call it a solution to alignment. It seems like “solving alignment” should involve figuring out how to make some sort of AI.
- We could say it is the problem of making an AI that makes things go well for a long time.
 - This seems too restrictive, because (at least a priori) there could be worlds in which there are lots of AIs, with no AI individually making things good forever, but with things still going well for a long time.
- **Maybe:** it is the problem of making AGIs that we could trust the future to.
 - But (at least a priori) maybe the future doesn't need to be “trusted to” AGIs? Maybe things could stay meaningfully under artificially enhanced human control?
- **Maybe:** it is the problem of figuring out how to make the AGI transition go well.

Should we speak of an “alignment endeavor” instead?

Notice a funny thing about some Anthropic/Christiano-style visions of the future:

- In these plans, when we speak of us “solving alignment”, we often mean making some initial AGIs that things can be handed off to, and then we imagine these AGIs “solving alignment” for their successors (which are potentially improved versions of the same AIs).
- This could repeat many times. Maybe it could repeat indefinitely?
- In case the future could be like this but still be good, we probably shouldn’t speak of “finding a solution to alignment”. I’d suggest “working on the alignment endeavor”.
- This is maybe like how the US founding fathers did not solve the problem democratic governance, but set in motion a thing that would hopefully keep solving problems of democratic governance.

This contrasts with MIRI-style views of alignment, where I think it’s more like some sort of recipe of an aligned ASI is found at some point.

An issue with viewing alignment as a continued endeavor

I think there is appeal to this view:

- Wouldn't it be weird if there were some grand formula of an artifact that makes things good forever after? It sounds like a device specification from an alchemist's imagination. Maybe there's no grand formula to be found, and humanity will instead need to continue to muddle through becoming more and more intelligent, like we always have muddled through everything.

But simultaneously I think holding this view makes it easy to make the following inference:

- There's no grand solution to be found, you just have to muddle through as always. And we will probably be kinda fine muddling through as always.

I think this is a bad inference. If anything, if it turns out to not be possible to find some sort of aligned ASI formula, that should maybe make you somewhat more pessimistic, because a type of path to good futures would be ruled out? And a civilization just muddling through does not go well for amoebas by default, and humans might well end up in the position of amoebas.

What we will say the alignment problem is, for this talk

- However, for the rest of this talk, I will set aside the possibility of there being some sort of long continued and potentially multipolar alignment mess. I will focus on the following MIRI-style version of the alignment problem:
 - How would one make an AI that makes things good for a long time?
- This is the focus mostly to make discussion easier.
- But I think it wouldn't be crazy for things to go this way.
- And anyway, even if getting a good AI future in practice does not look like this, the thinking you do about this is still very relevant.
- I should note that this basically involves taking over the world, which is pro tanto immoral. This aspect is worth discussing but outside the scope here.

Problems that alignment **reduces** to

- There are some problems such that if you solve them, you have solved alignment or can easily solve alignment.
- Said another way: solving alignment can be reduced to solving some other problems.
- We could call these **alignment-hard problems**.
 - this name comes from the fact that these problems are in a sense at least as hard as alignment
 - It is possible for an alignment-hard problem to be strictly harder than alignment.
- This is in analogy with with e.g. the term “NP-hard” from complexity theory. A problem is NP-hard when any problem in NP can be quickly reduced to that problem.
 - some examples of NP-hard problems: satisfiability, traveling salesman, TQBF, halting
 - That said, our situation is a bit disanalogous because it’s not clear what “the alignment complexity class” should mean.

Some canonical problems that AI alignment reduces to

I will quickly discuss the following canonical alignment-hard problems:

- Making an ASI sovereign with the right values
- Making a corrigible ASI
- Making a task ASI / genie
- Making an oracle
- Making some other sort of nice respectful ASI
- Making a (maybe more narrow or misaligned AI) that solves one of the above problems or enables some other pivotal act

The above list isn't exhaustive or mutually exclusive.

Value-aligned ASI sovereign

- You have an ASI that just has good values somehow “already written inside”
- Like, if you’re a kind of utilitarian, I guess this looks like an ASI just turning the universe into hedonium or many copies of some specific kind of “optimal-welfare being”.
- If you’re a utilitarian but slightly less “crazy”, maybe it looks like making a lot of humans with fulfilling lives. Maybe you make many copies of current Earth.
- I find it more confusing to think about what this is supposed to look like when you are less of a utilitarian or even if you care specifically about the specific humans who are currently alive.
 - Maybe it looks like just being very nice and helping all the humans?
 - Maybe it looks like running the ideal reflection process of each human to some satisfactory state and then doing with some fraction of the universe’s resources what they would want?

Corrigible AGI

- “A ‘corrigible’ agent is one that doesn’t interfere with what we would intuitively see as attempts to ‘correct’ the agent, or ‘correct’ our mistakes in building it; and permits these ‘corrections’”
- “an agent which, in an intuitive sense, reasons internally as if from the programmers' external perspective”
- An AGI corrigible to a human is supposed to be an AI system that ongoingly gets action-guidance etc from their values and extends their understanding.
 - The human-AI system could be like an enhanced version of the human.

Genies and oracles

- A genie, ie task AGI, is an AI that solves various tasks (and does nothing else).
- E.g. solves Yudkowsky's strawberry problem: "Place, onto this particular plate here, two strawberries identical down to the cellular but not molecular level."
- If you do this, at least if your genie is capable enough compared to the rest of the world, you can do a pivotal act such as:
 - asking it to enforce some sort of AGI ban for 100 years while otherwise disrupting human society minimally
 - asking it to build a nice ASI sovereign
- An oracle is a genie that only answers questions and solves other intellectual problems.
 - Specifically, maybe it only has a text output channel which is then read by humans.
 - The intention would be to get it to e.g. write a good alignment textbook, or a book that convinces the world to ban AGI, or to output safe ASI sovereign (training) code.

Making some other sort of nice respectful ASI

- Maybe you could make an ASI that isn't really corrigible to anyone and has lots of its own values/interests and does open-ended stuff, but that doesn't kill humans and puts some effort into protecting and helping humans so there is like at least a galaxy-wide human utopia?
- I think this is somewhat less bizarre than the previous options. (I think it's still somewhat bizarre though.)

Making a (narrow or misaligned) AI that helps solve one of these problems

- Arguably the version of this with a potentially misaligned AGI is the main practical plan of the current alignment field.

A couple more specific alignment-hard problems

I'll now talk about a couple more specific alignment-hard problems that I like.

Human prediction/imitation from counterfactual situations

- Maybe one could make a system that does stuff in a human-like way.
- Specifically, one can try to do this via very good human imitation or prediction.
- Pretrained language models are arguably an attempt at this.
 - [Conditioning Predictive Models](#) proposes still seeing RLHF-ed models this way.
 - The persona selection view tries to roughly see current LLMs this way.
- One could instead use ML methods to predict “finer” time series data from humans, like chains of thought or brain scans.
 - In the limit of reading human minds very precisely and just implementing that on a computer, one gets to mind uploads.
- In the broader sense of “make an AI that is kinda nice like a human”, I think Steven Byrnes is also working on this. His plan does not involve learning via imitation, it instead involves making synthetic human-like newborns that would need to grow up into nice humans in some virtual world.

Making an AI who makes and keeps a promise to be nice

- Maybe there's some way to make an AI that keeps its promises, ie if it says it will do something, it will do it, even across a lot of development, at least in "fair conditions"?
- And then maybe one can put it into a situation where it wants to make a promise to be nice to humans?
- E.g. you can hope to make a deal to split the universe with a misaligned AI who would keep its promises.
- See my post [Honorable AI](#) for a specific proposal of this kind.
 - Btw this is the best concrete reason I know of to think alignment is solvable (with some caveats given in the post).

Relaxations of the alignment problem

- A relaxation of the alignment problem is a version with some constraints relaxed/lifted. Some examples:
 - you could assume you have a practically unrealistic amount of compute
 - maybe you have a computer on which any program taking arbitrarily many steps magically finishes running in 1 sec
 - eg now you can try to run a vast number of simulated evolutions and find nice beings inside
 - maybe you additionally have an oracle answering halting questions
 - eg now you can approximate solomonoff induction
 - I will use “you have a hypercomputer” to loosely mean a strong version of this
 - you could assume you have a practically unrealistic amount of training data
 - maybe you have $10^{\{100\}}$ transcripts from Einstein spending a month solving a problem
 - you could assume you have practically inaccessible kinds of training data
 - maybe you have very detailed brain scans of lots of humans solving problems
 - maybe you have $10^{\{100\}}$ transcripts from Einstein solving a problem starting from the exact same mental+world state except for what is written on a piece of paper

Relaxations of the alignment problem (pt 2)

- But also we will just use “a relaxation of the alignment problem” in a less strict sense, to just mean a simpler/toy version of the alignment problem. Eg:
 - instead of having to help humans forever, you have to do something more concrete like making a lot of diamond or proving some formal mathematical statement (without wasting a lot of time)
 - we could assume our universe has some true fundamental laws of physics and these are known
 - we could assume the universe is some deterministic cellular automaton world
 - we could have an agent inserted into a video game world in a Cartesian way
 - we could magically have the entire world state be precisely known to the agent
- I recommend the learning/research strategy of trying to solve the simplest version that still seems hard

The diamond maximizer problem

- The diamond maximizer problem is the problem of, given an extremely large physical computer with some generic sensors and actuators (like, imagine it controls a bunch of humanoid robots, say), specifying a program to run on it which leads to there being a lot of carbon atoms which are covalently bound to four other carbon atoms.
 - This should be easier than alignment because “make diamond” is less confusing than “be very nice to humans”. It should also be easier because we’re granting a lot of compute.
 - The goal is to make almost as much diamond as possible given the resource constraints of our universe.
- You’re allowed to first use humans to gather a bunch of data
- There is also an “unreflective version”, in which there is somehow a Cartesian robot inside our universe which is hooked up to an arbitrary amount of compute outside.
 - This should be easier still because it alleviates some things with [embedded agency](#) issues.
- AFAIK even the unreflective version is an open problem.
- See these Arbital pages by Yudkowsky on this:
 - [Diamond maximizer](#)
 - [Ontology identification problem](#)

A first failed diamond maximizer solution attempt

Here's a bad diamond maximizer solution attempt, to communicate some of the difficulty.

- First, make a bunch of videos of the world, including of diamonds and the actuators.
- Then, world modeling / science / understanding is kinda about good prediction, right? So maybe find a short program that predicts the next frame of each video well from all previous frames — maybe this will be a good world model?
 - This can be made precise with Solomonoff induction.
- Ok, now we'd want to hook this “world model” up to some planning loop that finds actions which the world model says will lead to a lot of diamond existing. If it's a good world model, it should let us ask this question, right?
- Oops, the world model looks like this: “00011101010010101111...01001000”. Wtf does it mean to ask it about what actions would make a lot of diamond??

Exercise 1

Try to come up with a pair of simplified versions of the alignment problem such that:

- For the first version, you have a solution you're pretty confident would work.
 - Write down a specification of the problem and a sketch of your solution.
 - In case you're pretty confident you can solve even general versions of alignment, then the criterion is that you should be able to convince a fairly pessimistic friend that your solution works.
- The second version is only one or a few “hyperparameter changes” away from the first one, but you don't have a convincing solution to it.
 - Write down a specification of the problem. Try to write down intuitively what it is that is making it hard to solve, and speculate on what kind of understanding would help you solve it.
 - Again, if you're pretty confident you can solve even general versions of alignment, then the criterion is that you don't yet have a solution such that you could convince that pessimistic friend that your solution works.

(And then you can keep the second problem in mind for later as a longer-term challenge.)

The “research papers given hypercomputer” problem

- The problem: specify a program/system that gives a blueprint of a 1000x clock speed mind upload device (that doesn't also do something bad). Or a program/system that just gives math papers like human ones.
- You're given a hypercomputer, access to arbitrarily much data of the kinds that "already exist" or are reasonably easy per datum to gather, and ability to use a vast quantity of current humans to do fairly short monitoring/verification tasks.
 - Like, existing papers are completely fine to use as data, doing some easy reformatting/labeling of existing data is fine, having humans spend a few hours generating or evaluating stuff is fine, etc..
 - But having humanity think 1000 years and make mind uploads and then training a model which memorizes that and "can now make mind uploads" is not fine.
 - We assume that you're granted a completely secure/crisp server, ie one that only interacts with the outside world by receiving whatever inputs/data/code you give it, and acting on the external world only via the output channels you intend it to have (tho note that if you're monitoring it, then you might need to worry about whatever system you have monitoring it potentially getting used by it as an output channel).
- The point here is to specify a plan that clearly works (as opposed to a thing you might merely assign >50% chance of working after thinking about it for a few hours but still feel really confused about) — you should make a compelling case that your thing works.

Trying to solve the research|hypercomputer problem?

- I've spent a bunch of time thinking about how one would do this with predictive modeling.
- I have various things which I'm pretty sure do not work, and a few things such that I'm not completely convinced they don't work, but outside-view suspect they don't work.
- Here's an example imo-probably-failed solution attempt to a toy version:
 - Take all the human math papers, extract their main theorem statements and their proofs. Maybe formalize all of these.
 - Interleave the theorem statements and proofs into one string like $\text{thm } 1^* \text{ pf } 1^* \text{ thm } 2^* \text{ pf } 2^* \dots \text{ thm } n^* \text{ pf } n^*$, and then put the statement of the riemann hypothesis next as $\text{thm } n+1$. Throw this into an ideal prediction algorithm (specifically, I mean solomonoff induction), and have it predict how the sequence continues. Let's assume the riemann hypothesis is in fact true and provable.
 - Is what follows according to the predictor probably a proof of the riemann hypothesis?
- Exercise 2.1: Try to figure out if this would work or not.
- Exercise 2.2 (hard): Try to solve the problem.